Test of the overarrows package with all options

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1 Loading the package with many options

```
\usepackage[%
  old-arrows, esvecth, esvecta,
  tikz, pstricks, pstarrows,
  subscripts, allcommands, debug
]{overarrows}
```

2 Options `old-arrows` and `allcommands`

```
\TestOverArrow*{\overrightarrow}
```

Test of `\overrightarrow` and `\overrightarrow*` macros

`\overrightarrow` for different math styles

`\displaystyle`	`\textstyle`	`\scriptstyle`	`\scriptscriptstyle`
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

`\overrightarrow` kerning

$$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$$

`\overrightarrow*` kerning

$$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$$

```
\NewOverArrowCommand{\amsvec}{amsmath, end={\rightarrow}, shift left=2}
\TestOverArrow*{\amsvec}
```

Test of \amsvec and \amsvec* macros

\amsvec for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\amsvec kerning

$$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$$

\amsvec* kerning

$$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$$

3 Option esveca

```
\NewOverArrowCommand{\esvec}{esvect}
\TestOverArrow*{\esvec}
```

Test of \esvec and \esvec* macros

\esvec for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\vec{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\esvec kerning

$$\vec{t}_{\vec{u}\,\vec{v}} \qquad \vec{\imath}_0 \qquad \vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\,\vec{\imath} + v_y\,\vec{\jmath} + v_z\,\vec{k}$$

\esvec* kerning

$$\vec{t}_{\vec{u}\,\vec{v}} \qquad \vec{\imath}_0 \qquad \vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\,\vec{\imath} + v_y\,\vec{\jmath} + v_z\,\vec{k}$$

4 Option `tikz`

```
\NewOverArrowCommand[tikz]{\tikzvec}{}
\TestOverArrow*{\tikzvec}
```

Test of \tikzvec and \tikzvec* macros

\tikzvec for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\vec{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\tikzvec kerning

$\vec{t}_{\vec{u}_{\vec{v}}} \quad \vec{\imath}_0 \quad \vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\,\vec{\imath} + v_y\,\vec{\jmath} + v_z\,\vec{k}$

\tikzvec* kerning

$\vec{t}_{\vec{u}_{\vec{v}}} \quad \vec{\imath}_0 \quad \vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\,\vec{\imath} + v_y\,\vec{\jmath} + v_z\,\vec{k}$

```
\NewOverArrowCommand[tikz]{\tikzvec}{}
\NewOverArrowCommand[tikz]{\thinnertikzvec}{%
  thinner,
}
\NewOverArrowCommand[tikz]{\thickertikzvec}{%
  line thickness={2\overarrowthickness},
}

$$ \thinnertikzvec{v} \qquad \tikzvec{v} \qquad \thickertikzvec{v} $$
```

$\vec{v} \qquad \vec{v} \qquad \vec{v}$

```
\NewOverArrowCommand[tikz]{\tikzoptvec}{%
  tikz options={line width=2\overarrowthickness},
  path options={arrows={<->}},
  path={(0,0)--(0.5,0.05)},
}
\TestOverArrow*{\tikzoptvec}
```

Test of \tikzoptvec and \tikzoptvec* macros

\tikzoptvec for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overleftrightarrow{v}$	$\overleftrightarrow{v}$	$\overleftrightarrow{v}$	$\overleftrightarrow{v}$
$\overleftrightarrow{AB}$	$\overleftrightarrow{AB}$	$\overleftrightarrow{AB}$	$\overleftrightarrow{AB}$
$\overleftrightarrow{\text{grad}}$	$\overleftrightarrow{\text{grad}}$	$\overleftrightarrow{\text{grad}}$	$\overleftrightarrow{\text{grad}}$
$\overleftrightarrow{my\ long\ vector}$	$\overleftrightarrow{my\ long\ vector}$	$\overleftrightarrow{my\ long\ vector}$	$\overleftrightarrow{my\ long\ vector}$

\tikzoptvec kerning

$\overleftrightarrow{t}_{\overleftrightarrow{u}_{\overleftrightarrow{v}}}$ $\overleftrightarrow{\imath_0}$ $\overleftrightarrow{v} = \overleftrightarrow{v_x} + \overleftrightarrow{v_y} + \overleftrightarrow{v_z} = v_x\overleftrightarrow{\imath} + v_y\overleftrightarrow{\jmath} + v_z\overleftrightarrow{k}$

\tikzoptvec* kerning

$\overleftrightarrow{t}_{\overleftrightarrow{u}_{\overleftrightarrow{v}}}$ $\overleftrightarrow{\imath_0}$ $\overleftrightarrow{v} = \overleftrightarrow{v_x} + \overleftrightarrow{v_y} + \overleftrightarrow{v_z} = v_x\overleftrightarrow{\imath} + v_y\overleftrightarrow{\jmath} + v_z\overleftrightarrow{k}$

```
\NewOverArrowCommand[tikz]{\tikzaddoptvec}{%
  add tikz options={blue},
  add path options={thick},
  arrows={->>}, min length=20,
}
\TestOverArrow*{\tikzaddoptvec}
```

Test of \tikzaddoptvec and \tikzaddoptvec* macros

\tikzaddoptvec for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{\mathit{my\ long\ vector}}$	$\overrightarrow{\mathit{my\ long\ vector}}$	$\overrightarrow{\mathit{my\ long\ vector}}$	$\overrightarrow{\mathit{my\ long\ vector}}$

\tikzaddoptvec kerning

$\overrightarrow{t}_{\overrightarrow{u}}^{\overrightarrow{v}} \qquad \overrightarrow{l_0} \qquad \overrightarrow{v} = \overrightarrow{v_x} + \overrightarrow{v_y} + \overrightarrow{v_z} = v_x\,\overrightarrow{i} + v_y\,\overrightarrow{j} + v_z\,\overrightarrow{k}$

\tikzaddoptvec* kerning

$\overrightarrow{t}_{\overrightarrow{u}}^{\overrightarrow{v}} \qquad \overrightarrow{l_0} \qquad \overrightarrow{v} = \overrightarrow{v_x} + \overrightarrow{v_y} + \overrightarrow{v_z} = v_x\,\overrightarrow{i} + v_y\,\overrightarrow{j} + v_z\,\overrightarrow{k}$

```
\NewOverArrowCommand[tikz]{\tikcodevec}{%
  tikz command={\draw[Circle-Circle](0,0).. controls (0.5,0.3).. (1,0);},
  shift left=-2, space after arrow=0.2ex,
}
\TestOverArrow*{\tikcodevec}
```

Test of \tikcodevec and \tikcodevec* macros

\tikcodevec for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\vec{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$
$\vec{AB}$	$\vec{AB}$	$\vec{AB}$	$\vec{AB}$
$\vec{\text{grad}}$	$\vec{\text{grad}}$	$\vec{\text{grad}}$	$\vec{\text{grad}}$
$\vec{\textit{my long vector}}$	$\vec{\textit{my long vector}}$	$\vec{\textit{my long vector}}$	$\vec{\textit{my long vector}}$

\tikcodevec kerning

$\vec{t}_{\vec{u}_{\vec{v}}}$ $\vec{i}_0$ $\vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\vec{i} + v_y\vec{j} + v_z\vec{k}$

\tikcodevec* kerning

$\vec{t}_{\vec{u}_{\vec{v}}}$ $\vec{i}_0$ $\vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\vec{i} + v_y\vec{j} + v_z\vec{k}$

5 Option pstricks

```
\NewOverArrowCommand[pstricks]{\pstvec}{}
\TestOverArrow*{\pstvec}
```

Test of \pstvec and \pstvec* macros

\pstvec for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\vec{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\pstvec kerning

$$\vec{t}_{\vec{u}_{\vec{v}}} \qquad \vec{\imath}_0 \qquad \vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\vec{\imath} + v_y\vec{\jmath} + v_z\vec{k}$$

\pstvec* kerning

$$\vec{t}_{\vec{u}_{\vec{v}}} \qquad \vec{\imath}_0 \qquad \vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\vec{\imath} + v_y\vec{\jmath} + v_z\vec{k}$$

```
\NewOverArrowCommand[pstricks]{\pstTbarandarrows}{arrow={|<->|}, center arrow}
\TestOverArrow*{\pstTbarandarrows}
```

Test of \pstTbarandarrows and \pstTbarandarrows* macros

\pstTbarandarrows for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overarrow{v}$	$\overarrow{v}$	$\overarrow{v}$	$\overarrow{v}$
$\overarrow{AB}$	$\overarrow{AB}$	$\overarrow{AB}$	$\overarrow{AB}$
$\overarrow{\mathrm{grad}}$	$\overarrow{\mathrm{grad}}$	$\overarrow{\mathrm{grad}}$	$\overarrow{\mathrm{grad}}$
$\overarrow{my\ long\ vector}$	$\overarrow{my\ long\ vector}$	$\overarrow{my\ long\ vector}$	$\overarrow{my\ long\ vector}$

\pstTbarandarrows kerning

$$\overarrow{t_{\overarrow{u}_{\overarrow{v}}}} \qquad \overarrow{i}_0 \qquad \overarrow{v} = \overarrow{v}_x + \overarrow{v}_y + \overarrow{v}_z = v_x\overarrow{i} + v_y\overarrow{j} + v_z\overarrow{k}$$

\pstTbarandarrows* kerning

$$\overarrow{t_{\overarrow{u}_{\overarrow{v}}}} \qquad \overarrow{i}_0 \qquad \overarrow{v} = \overarrow{v}_x + \overarrow{v}_y + \overarrow{v}_z = v_x\overarrow{i} + v_y\overarrow{j} + v_z\overarrow{k}$$

```
\NewOverArrowCommand[pstricks]{\pstsmallbluearrow}{
  psset = {arrowscale=0.5, arrowinset=0, linecolor=blue},
  thinner,
}
\TestOverArrow*{\pstsmallbluearrow}
```

Test of \pstsmallbluearrow and \pstsmallbluearrow* macros

\pstsmallbluearrow for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\pstsmallbluearrow kerning

$$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\overrightarrow{\imath} + v_y\overrightarrow{\jmath} + v_z\overrightarrow{k}$$

\pstsmallbluearrow* kerning

$$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\overrightarrow{\imath} + v_y\overrightarrow{\jmath} + v_z\overrightarrow{k}$$

```
\NewOverArrowCommand[pstricks]{\psellipticarrow}{
  pstricks command={%
    \psellipticarcn{->}%
    (0.5\overarrowlength,0.2\overarrowlength)%
    (0.5\overarrowlength,0.2\overarrowlength)%
    {170}{10}
  },
  geometry={(0,0.2\overarrowlength)(\overarrowlength,0.4\overarrowlength)},
  center arrow,
}
\TestOverArrow*{\psellipticarrow}
```

Test of \psellipticarrow and \psellipticarrow* macros

\psellipticarrow for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\psellipticarrow kerning

$$\vec{t}_{\vec{u}_{\vec{v}}} \qquad \vec{t}_0 \qquad \vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\vec{\imath} + v_y\vec{\jmath} + v_z\vec{k}$$

\psellipticarrow* kerning

$$\vec{t}_{\vec{u}_{\vec{v}}} \qquad \vec{t}_0 \qquad \vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\vec{\imath} + v_y\vec{\jmath} + v_z\vec{k}$$

6 Option pstarrows

```
\NewOverArrowCommand[picture]{\picvec}{}
\TestOverArrow*{\picvec}
```

Test of \picvec and \picvec* macros

\picvec for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\picvec kerning

$$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$$

\picvec* kerning

$$\overrightarrow{t}_{\overrightarrow{u}_{\overrightarrow{v}}} \qquad \overrightarrow{\imath}_0 \qquad \overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$$

7 Option subscripts

```
\NewOverArrowCommand{\subvec}{min length=30}
\NewOverArrowCommand{\nosubvec}{min length=30, detect subscripts=false}
\TestOverArrow*{\subvec}
\TestOverArrow*{\nosubvec}
```

Test of \subvec and \subvec* macros

\subvec for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\subvec kerning

$\overrightarrow{t}_{\overrightarrow{u}\,\overrightarrow{v}}$ $\overrightarrow{\imath_0}$ $\overrightarrow{v} = \overrightarrow{v_x} + \overrightarrow{v_y} + \overrightarrow{v_z} = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$

\subvec* kerning

$\overrightarrow{t}_{\overrightarrow{u}\,\overrightarrow{v}}$ $\overrightarrow{\imath_0}$ $\overrightarrow{v} = \overrightarrow{v_x} + \overrightarrow{v_y} + \overrightarrow{v_z} = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$

Test of \nosubvec and \nosubvec* macros

\nosubvec for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\nosubvec kerning

$\overrightarrow{t}_{\overrightarrow{u}\,\overrightarrow{v}}$ $\overrightarrow{\imath}_0$ $\overrightarrow{v} = \overrightarrow{v}_x + \overrightarrow{v}_y + \overrightarrow{v}_z = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$

\nosubvec* kerning

$\overrightarrow{t}_{\overrightarrow{u}\,\overrightarrow{v}}$ $\overrightarrow{\imath_0}$ $\overrightarrow{v} = \overrightarrow{v_x} + \overrightarrow{v_y} + \overrightarrow{v_z} = v_x\,\overrightarrow{\imath} + v_y\,\overrightarrow{\jmath} + v_z\,\overrightarrow{k}$